

WGCNA Analysis of mRNA Expression Data: Pipeline Structure, Results, and Signed vs. Unsigned Network Comparison

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Contents

1	Pipeline Structure	2
1.1	Global Parameters	2
1.2	Loading, Low-Expression Filtering, and Size-Factor Export	2
1.3	MAD-Based Feature Selection and Why MAD Over Variance	2
1.4	Sample Clustering (Outlier Screening)	3
1.5	Module–Trait Relationship Logic	3
1.6	Network Construction, GO Enrichment, and Hub Genes (brief)	3
2	Results	4
2.1	Cohort and Preprocessing Summary	4
2.2	MAD Distribution	4
2.3	Sample Clustering	4
2.4	Soft-Threshold Selection	5
2.5	Network Construction and Module Detection	6
2.6	Module Eigengene Correlation Structure	9
2.7	Module–Trait Relationships	10
2.8	Hub Genes	11
3	Reasons for Choosing Signed Network	12
4	GO Enrichment Analysis of the Signed Network	13
4.1	Turquoise: Cell Cycle / DNA Replication	14
4.2	Brown: Ciliated/Differentiated Epithelial Identity	15
4.3	Blue: Immune / Leukocyte Activation	16
4.4	Green: Ribosome Biogenesis / Oxidative Phosphorylation	18
4.5	Yellow: Vasculature / Endothelium	19
4.6	Red: Extracellular Matrix / Collagen	21
4.7	Pink: DNA Repair / Mitochondrial Envelope	22
4.8	Black: Ciliary Transition Zone	23
4.9	Magenta: Motile Cilia / Axonemal Dynein	24
4.10	Purple: Chromosome 8q24.3 Amplicon	25
4.11	Greenyellow: Histone Cluster / Chromatin Assembly	26
4.12	Summary	27

1 Pipeline Structure

1.1 Global Parameters

The script first fixes a small set of global parameters that control the entire run:

- `MIN_COUNT = 10`, `MIN_FRAC = 0.50`: the low-expression filter. A gene is retained only if it has at least 10 reads in at least 50% of the 963 samples. This is included to remove genes with negligible expression, not to do any variance-based curation, which will be performed by MAD further.
- `N_GENES_WGCNA = NA`: controls whether the gene set is truncated to the top- N most variable (by MAD) genes before WGCNA. Setting this to `NA` means *every* gene that survives the expression filter is carried into WGCNA; no additional hard cutoff is imposed.
- `MAX_BLOCK_SIZE = 12000`: raised well above the WGCNA default of 5000 so that all 8692 genes are processed as a single block.
- `N_HUBS = 20`: the number of top-kME genes exported per module for the hub-gene tables.

NOTE: All the filtering steps (`MIN_COUNT`, `MIN_FRAC`, `N_GENES_WGCNA`) are set as `NA`, as the data provided was already filtered, as confirmed in the meeting.

1.2 Loading, Low-Expression Filtering, and Size-Factor Export

The raw file (`mRNA_filter1.xlsx`) is read as a 963×8693 sample-by-column table, where one column is the PAM50 label (with “TNBC” relabelled to “Basal” to match standard nomenclature) and the remaining 8692 columns are raw counts. The matrix is transposed to the conventional genes-by-samples orientation for DESeq2.

DESeq2 is then used, but in a **restricted role**: the design formula is ~ 1 (an intercept-only model), so DESeq2 is *not* being used here to test for differential expression. It is used purely for normalisation, specifically to run `estimateSizeFactors()` (median-of-ratios normalisation, which corrects for differences in sequencing depth across the 963 samples) and `estimateDispersions()` (the mean–variance trend that the variance-stabilising transform needs). The resulting size factors are written out explicitly:

```
saveRDS(sizeFactors(dds_mrna),
        file.path(OUT_DIR, "mrna_size_factors.rds"))
```

This export exists so that the lncRNA pipeline can reuse the *same* size factors rather than re-estimating its own. ***The lncRNA count submatrix is far sparser (lower per-gene counts, more zeros) than the mRNA matrix***, so DESeq2’s size-factor estimator is less reliable if run on the lncRNA data in isolation (library depth is a property of the sequencing run, not of the transcript biotype), so it is more robust to estimate it once from the denser mRNA counts and carry it across. After normalisation, `varianceStabilizingTransformation()` (blind to the subtype labels) produces the VST matrix that all downstream steps operate on.

1.3 MAD-Based Feature Selection and Why MAD Over Variance

NOTE: This subsection is only for reference.

Why MAD Over Standard Variance for Feature Selection:

- **Outlier Sensitivity:** MAD is defined as $\text{median}(|x_i - \text{median}(x)|)$, whereas variance relies on squared deviations from the mean ($|x_i - \bar{x}|^2$). Squaring heavily penalizes large deviations, allowing a small number of extreme samples (e.g., mis-normalization, batch artifacts, or extreme biological outliers) to artificially inflate a gene’s variance score.

- **Cohort-Wide Consistency:** MAD selects genes based on consistent, reproducible expression variability across the whole dataset rather than variance driven by extreme values in a few samples. *It is also recommended by standard WGCNA tutorials.*

In this run, `N_GENES_WGCNA` was set to `NA`, so no hard MAD cutoff was actually applied. The histogram in Figure 1 was generated as a diagnostic (and the code contains a conditional to draw a vertical cutoff line on it, for use in future runs where a top- N truncation is desired).

1.4 Sample Clustering (Outlier Screening)

Before network construction, samples are hierarchically clustered (average linkage on Euclidean distance across all 8692 genes) purely as a visual outlier check:

```
sampleTree_mrna <- hclust(dist(datExpr_mrna), method = "average")
plot(sampleTree_mrna, ...)
CUT_HEIGHT_MRNA <- NA # set to a numeric height to actually cut
```

The cut height, for this run, is left as `NA` in the script. The dendrogram is plotted and inspected first, and only if a genuinely isolated cluster of arrays is visible at some height would that height be filled in and a `cutreeStatic()` call used to drop those samples. No such height was set in this run, so all 963 samples that passed `goodSamplesGenes()` were retained.

1.5 Module–Trait Relationship Logic

PAM50 subtype is a four-level categorical variable, so it cannot be correlated with a module eigengene directly. The script converts it into a one-hot trait matrix:

This produces four binary indicator columns (Basal, HER2, LumA, LumB), each a 0/1 column across the 963 samples.

For each module, the eigengene (first principal component of that module’s expression sub-matrix, one value per sample) is correlated against each of the four indicator columns using Pearson correlation, and the corresponding p -value is obtained from `corPvalueStudent()`, which converts a correlation coefficient and sample size into a Student’s t -based p -value under the null hypothesis of no association. *This is repeated once for the signed network’s eigengenes and once for the unsigned network’s, producing the two module–trait heatmaps (Figures 8 and 9) and their corresponding CSV tables.*

1.6 Network Construction, GO Enrichment, and Hub Genes (brief)

Soft-thresholding power is chosen automatically via `pickSoftThreshold()` across candidate powers $\{1, \dots, 10, 12, 14, 16, 18, 20\}$, separately for signed and unsigned adjacency, taking the lowest power at which the scale-free topology fit R^2 first exceeds 0.90 (falling back to a manual default if no power reaches that target). `blockwiseModules()` was then run twice for each network type: the first run (`deepSplit=2`, `minModuleSize=30`) required some hyperparameter tuning, (`deepSplit=1`, `minModuleSize=50`, same `mergeCutHeight=0.25`). The tuned version actually carried forward into every downstream analysis (module colours, eigengenes, module–trait correlations, GO enrichment, hub genes).

2 Results

2.1 Cohort and Preprocessing Summary

The starting cohort comprised 963 tumour samples with PAM50 subtype calls distributed as shown in Table 1. Class imbalance (LumA making up more than half the cohort, HER2 well under 10%) is observed.

Table 1: PAM50 subtype composition of the 963-sample cohort.

	Basal	HER2	LumA	LumB
Samples	170	75	528	190

Library sizes ranged from 7.3M to 45.7M reads (median 22.7M), and DESeq2 size factors ranged from 0.374 to 2.048 (median 1.029), indicating a roughly $5.5\times$ spread in effective sequencing depth across the cohort that the size-factor normalisation and subsequent VST are correcting for.

2.2 MAD Distribution

Figure 1 shows the distribution of gene-level MAD on the VST scale across all the genes. The distribution is strongly right-skewed: the bulk of genes sit in a fairly narrow band of low-to-moderate MAD (roughly 0.5–1.5), with a long thin tail extending out past MAD = 4–6 for a small number of highly variable genes. Because `N_GENES_WGCNA` was left at NA, no cut was drawn on this distribution for the present run.

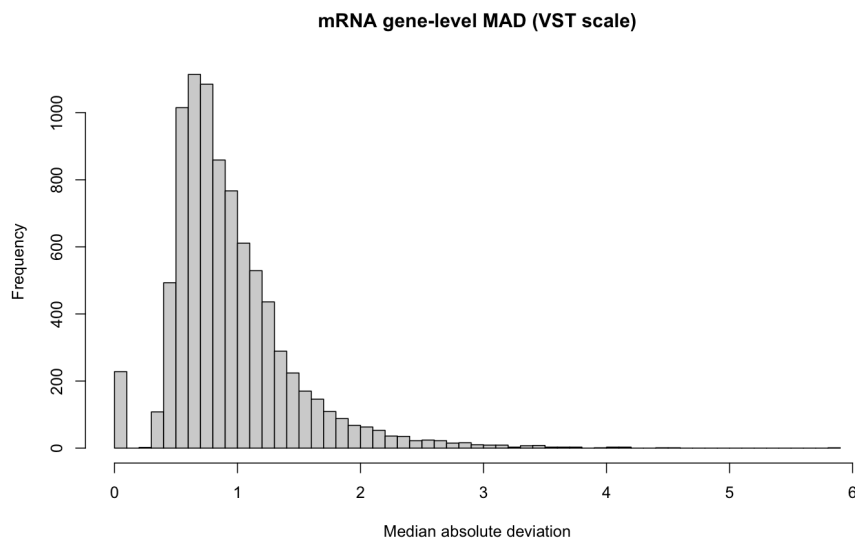


Figure 1: Distribution of gene-level MAD (VST scale) across all the given mRNA genes.

2.3 Sample Clustering

Figure 2 shows the average-linkage sample dendrogram used for outlier screening. There is no single sample or small sample cluster sitting at a dramatically greater height than the rest of the tree in a way that would indicate a technical failure; the tree instead shows the expected fine-grained substructure of a large, biologically heterogeneous tumour cohort. On this basis no cut height was applied and all 963 samples were retained.

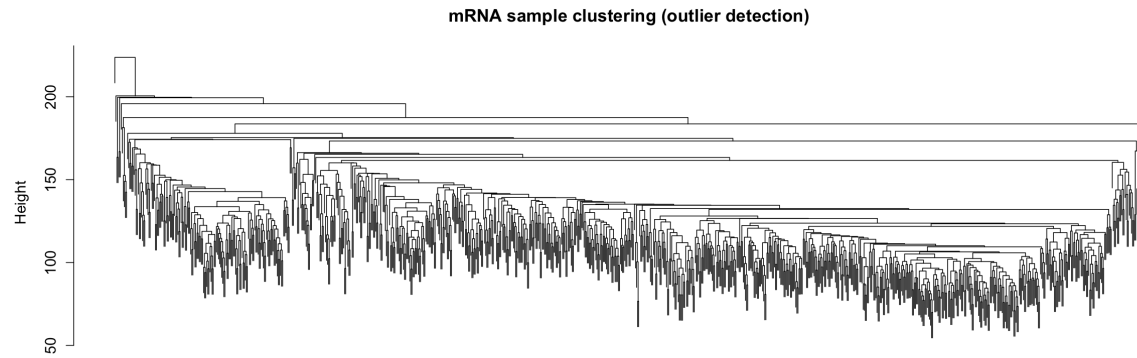


Figure 2: Average-linkage hierarchical clustering of all 963 samples (outlier screening dendrogram). No cut was applied.

2.4 Soft-Threshold Selection

Figure 3 shows the scale-free topology fit (R^2 , top row) and mean connectivity (bottom row) curves for both network types across the candidate power sequence.

Table 2: Selected soft-thresholding power and fit statistics for each network type.

Network type	Chosen power	Scale-free R^2 at chosen power	Mean connectivity
Unsigned	7	0.917	3.97
Signed	14	0.920	7.77

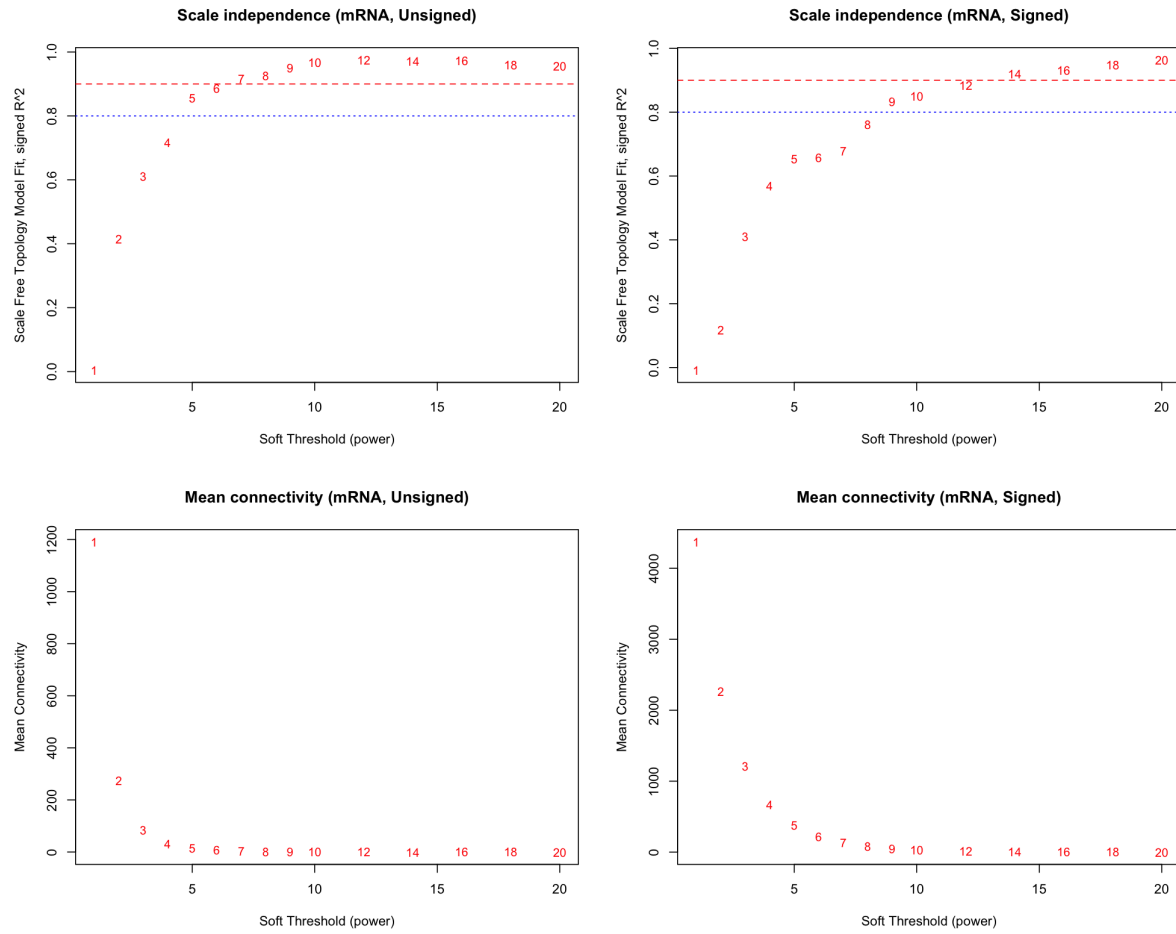


Figure 3: Scale-free topology fit and mean connectivity as a function of soft-thresholding power, for the unsigned (left column) and signed (right column) networks.

2.5 Network Construction and Module Detection

Table 3 reports the final ("tuned", `deepSplit=1`, `minModuleSize=50`) module sizes for both network types. The signed network resolves into 9 non-grey modules covering 4050 of 7220 genes (56.1%), while the unsigned network resolves into 7 non-grey modules covering only 2644 of 7220 genes (36.6%). *The unsigned grey module is nearly $1.5\times$ larger in gene count than the signed grey module (4576 vs. 3170 genes).*

Table 3: Final tuned module sizes (gene counts) for the signed and unsigned mRNA networks. Grey denotes unassigned genes.

Module	Signed	Unsigned
black	310	53
blue	699	596
brown	673	576
green	530	310
greenyellow	54	—
magenta	71	—
pink	268	53
purple	67	—
red	527	91
turquoise	793	607
yellow	541	427
grey (unassigned)	4159	5979
Total non-grey	4533	2713
Total genes	8692	8692

Figure 4 and 5 shows the gene clustering dendrograms with module colour bars for both networks (power 14 signed, power 7 unsigned).

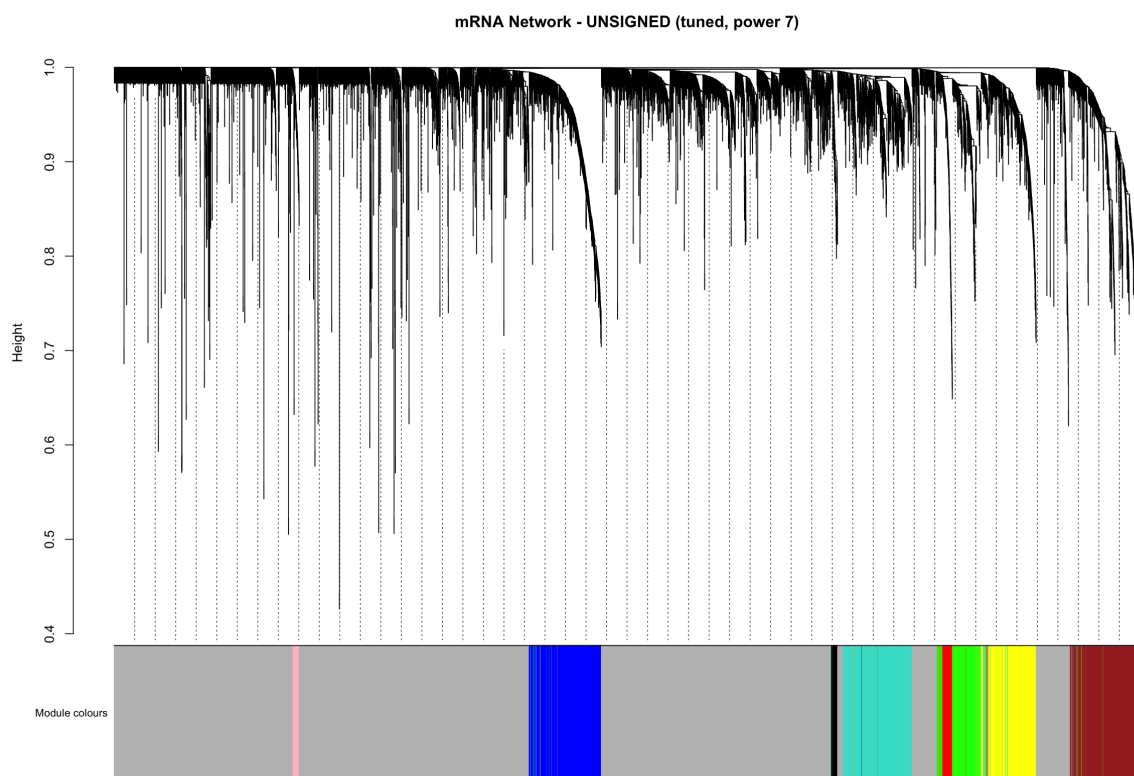


Figure 4: Gene clustering dendrogram with module colour assignment for the tuned unsigned mRNA network.

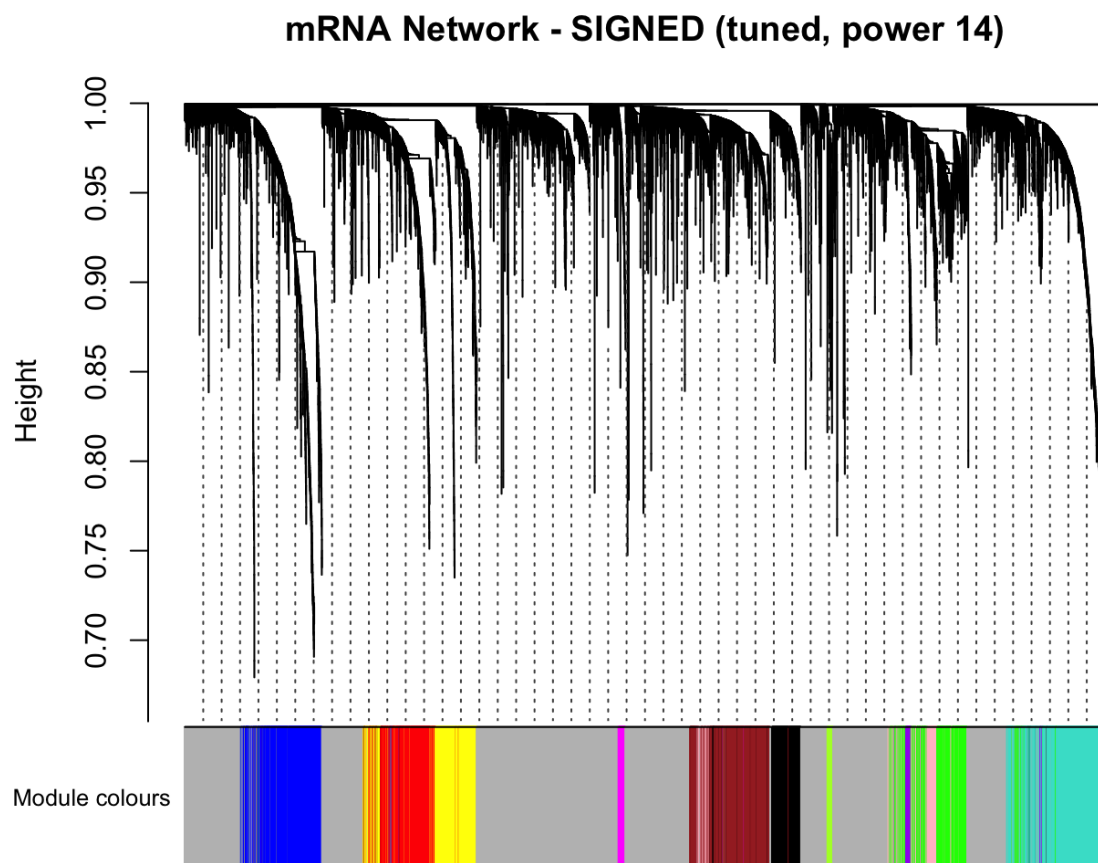


Figure 5: Gene clustering dendrogram with module colour assignment for the tuned signed mRNA network.

2.6 Module Eigengene Correlation Structure

Figure 6 and 7 shows the pairwise correlation of module eigengenes within both the networks.

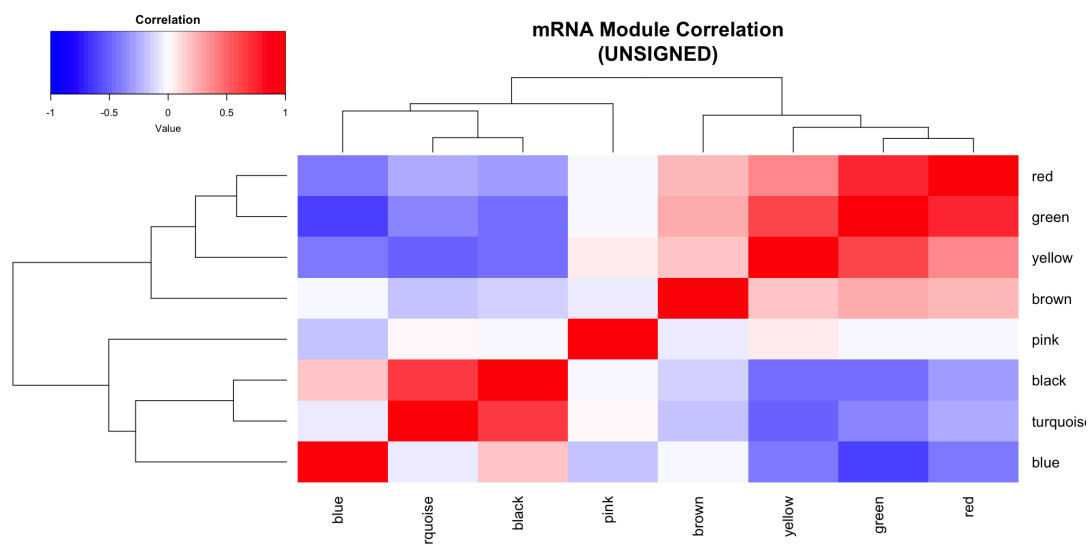


Figure 6: Module eigengene correlation heatmap for the unsigned mRNA network, with hierarchical clustering of modules by eigengene correlation.

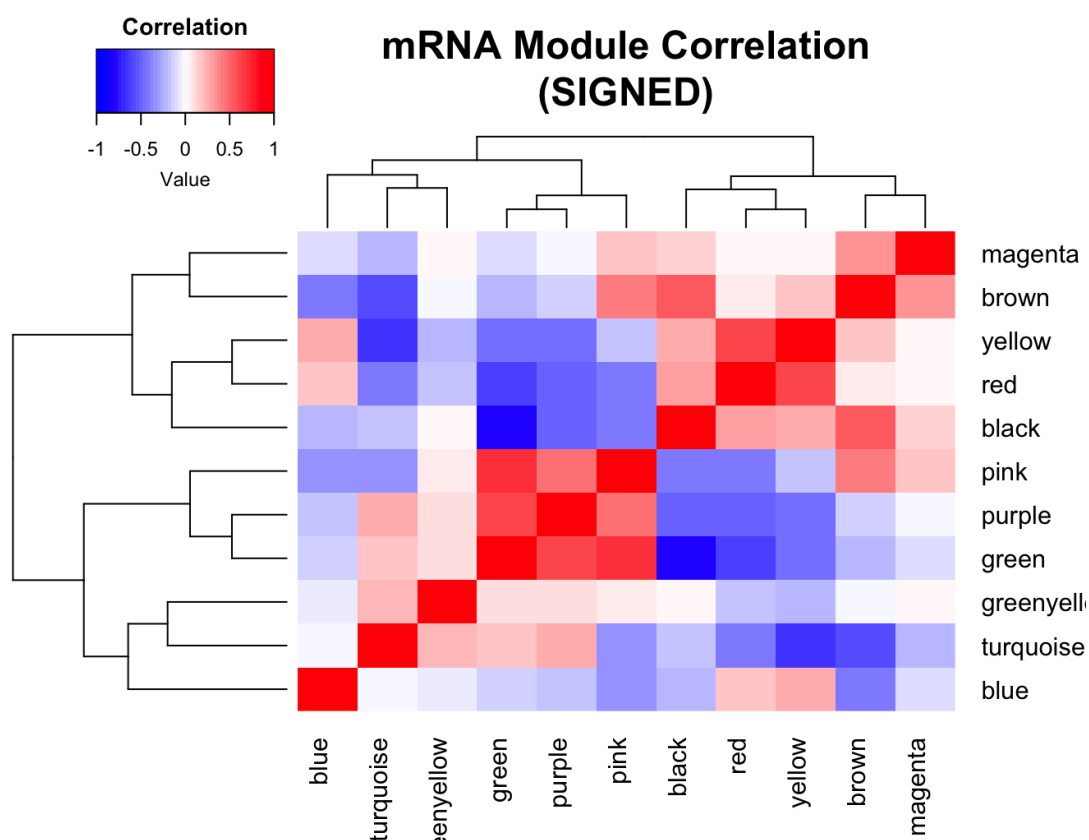


Figure 7: Module eigengene correlation heatmap for the signed mRNA network, with hierarchical clustering of modules by eigengene correlation.

2.7 Module–Trait Relationships

Figures 8 and 9 show the full module–subtype correlation heatmaps (Pearson r and Student p -value per cell) for the signed and unsigned networks respectively; the underlying values for the signed network are reproduced in Table 4.

Table 4: Signed module–subtype Pearson correlations (**n.s. marks $p > 0.05$; all other cells are significant, though several fall between 10^{-4} and 0.05 rather than below 10^{-4} .**)

Module	Basal	HER2	LumA	LumB
turquoise	0.56	0.14	−0.73	0.29
brown	−0.86	−0.16	0.61	0.17
black	−0.39	−0.15	0.34	0.05 (n.s.)
pink	−0.37	−0.01 (n.s.)	0.24	0.06 (n.s.)
magenta	−0.29	−0.06	0.18	0.09
yellow	−0.26	−0.10	0.51	−0.31
red	−0.20	0.005 (n.s.)	0.30	−0.19
blue	0.24	0.11	−0.14	−0.13
green	0.22	0.08	−0.23	0.02 (n.s.)
purple	0.15	−0.03 (n.s.)	−0.26	0.20
greenyellow	0.02 (n.s.)	0.11	−0.15	0.10

The dominant pattern across both heatmaps is a strong **Basal–LumA opposition**: turquoise and brown are the two largest-magnitude modules (brown : $r = -0.86$ with Basal, $r = 0.61$ with LumA; turquoise: $r = 0.56$ with Basal, $r = -0.73$ with LumA), and every other module’s correlation sign is consistent with belonging to one side or the other of this same axis. **HER2 is comparatively weakly represented across all modules** (largest $|r| = 0.16$), and several module–HER2 cells are not significant ($p > 0.05$), which matches the smaller HER2 sample size ($n=75$) and its known intermediate transcriptional profile relative to Basal and LumA.

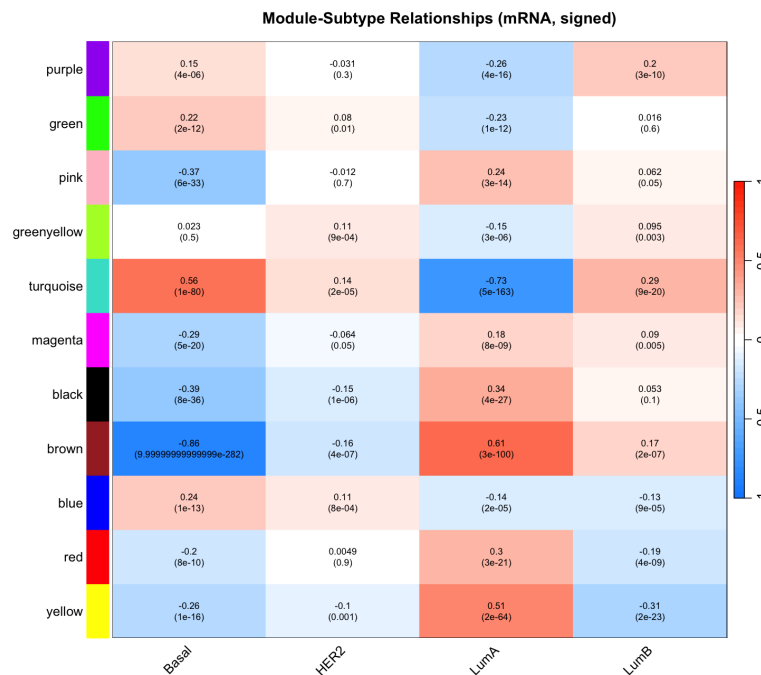


Figure 8: Module–subtype relationships, signed mRNA network. Each cell reports Pearson r (top) and p -value (bottom).

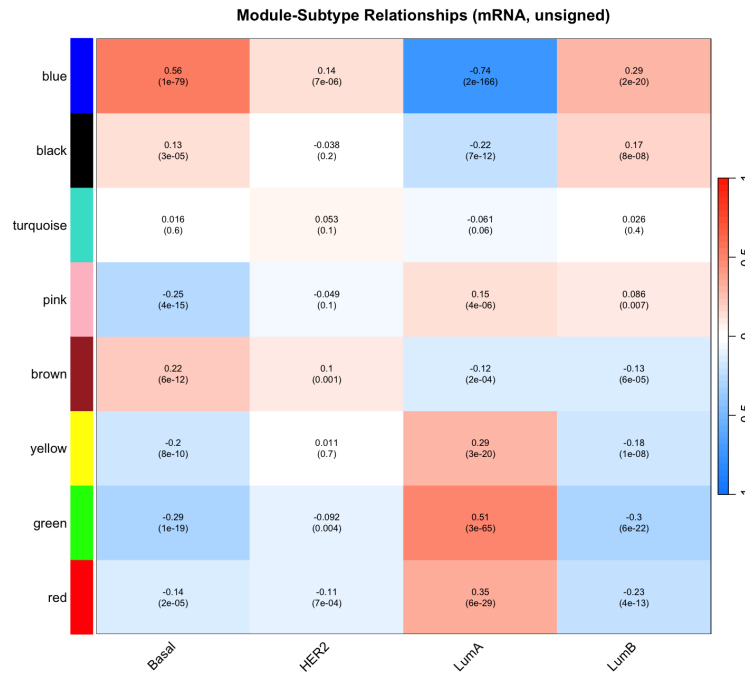


Figure 9: Module-subtype relationships, unsigned mRNA network. Each cell reports Pearson r (top) and p -value (bottom).

2.8 Hub Genes

Hub genes were extracted as the top 20 genes per module by signed kME, restricted to genes actually assigned to that module. The resulting table was then annotated through a script, which automatically queries MyGene.info for the full gene name, aliases and Entrez ID, and cBioPortal for the TCGA-BRCA somatic mutation and GISTIC-derived copy-number alteration frequency of each gene.

Table 5: Top kME hub gene per signed module, with its TCGA-BRCA alteration frequency. **All values from the unfiltered run.**

Module	Top hub	kME	Mutation %	CNA %	Altered %
blue	<i>SNX20</i>	0.958	0.1	1.8	1.8
red	<i>FBN1</i>	0.927	2.2	0.7	2.8
purple	<i>PUF60</i>	0.920	0.2	10.7	10.7
turquoise	<i>MCM10</i>	0.914	0.8	2.2	3.0
black	<i>DMXL1</i>	0.909	2.0	0.6	2.5
magenta	<i>APOBEC4</i>	0.904	0.0	7.4	7.3
pink	<i>MCRIP2</i>	0.893	0.1	3.9	4.0
brown	<i>ESR1</i>	0.886	0.8	2.2	3.0
yellow	<i>S1PR1</i>	0.886	0.1	0.5	0.6
greenyellow	<i>H4C4</i>	0.862	0.5	1.1	1.6
green	<i>NAA10</i>	0.858	0.3	1.6	1.8

The top six hubs of each module read as a coherent list in almost every case: **turquoise** gives *MCM10*, *TPX2*, *TTK*, *CDCA8*, *BUB1*, *CCNA2*, all mitotic; **brown** gives *ESR1*, *MLPH*, *FOXA1*, *TBC1D9*, *TTC6*, *THSD4*, all canonical luminal breast markers; **blue** gives *SNX20*, *PTPN7*, *IL12RB1*, *SASH3*, *CD53*, *CD48*, all leukocyte-restricted; **red** gives *FBN1*, *GLT8D2*,

COL6A3, *COL5A2*, *COL3A1*, *FSTL1*, all matrix and collagen; and **greenyellow** gives *H4C4*, *H2BC17*, *H2BC7*, *H4C5*, *H2AC13*, *H3C10*, which are all clustered histone genes.

An interesting find: Every one of the twenty **purple** hub genes carries a copy-number alteration frequency between 10.6% and 10.8%, an almost identical value, while their mutation frequencies are all below 1%. The members named, *PUF60*, *HSF1*, *RECQL4*, *SHARPIN*, *SLC52A2*, *FBXL6*, *GPAA1* and *SCRIB*, are all located in the **8q24.3 region, which is a recurrently amplified locus in breast cancer**.

The single highest-altered gene anywhere in the table is *GATA3* in **brown** at 14.6% altered, driven by an 11.9% mutation rate. *GATA3* is one of the most frequently mutated genes in luminal breast cancer, and it sits in the luminal module alongside *ESR1* and *FOXA1*, as expected.

3 Reasons for Choosing Signed Network

1. **Prevention of Signal Cancellation:** Unsigned adjacency uses absolute correlation values ($|r|$), grouping strongly positively and strongly negatively correlated genes into the same module. In this dataset, the **607**-gene unsigned **turquoise** module shows near-zero correlation across all PAM50 subtypes ($|r| \leq 0.061$, $p \geq 0.06$). When constructed signed, these same genes separate into the **brown** and **turquoise** modules, yielding strong subtype correlations ($|r|$ up to 0.86). Computing eigengenes from anti-correlated genes in the unsigned network causes expression signals to cancel out.
2. **Higher Coverage:** The signed network assigned 52.2% (4,533/8,692) of genes to non-grey modules, compared to 31.2% (2,713/8,692) in the unsigned network under identical parameters (`minModuleSize` and `mergeCutHeight`).

4 GO Enrichment Analysis of the Signed Network

GO over-representation analysis (`clusterProfiler::enrichGO()`, ontologies BP/CC/MF, background = all **8692** genes entering WGCNA, BH-adjusted $p < 0.05$ and $q < 0.05$) was run for all **eleven** non-grey signed modules. Every module returned significant terms in at least one ontology. The combined BP/CC/MF dot plots are shown below (top 10 terms per ontology, ranked by gene ratio), one subsection per module, together with a short interpretation that cross-references each module's dominant subtype correlation (Section 2.7, Table 4) and its top kME hub gene (Section 2.8).

4.1 Turquoise: Cell Cycle / DNA Replication

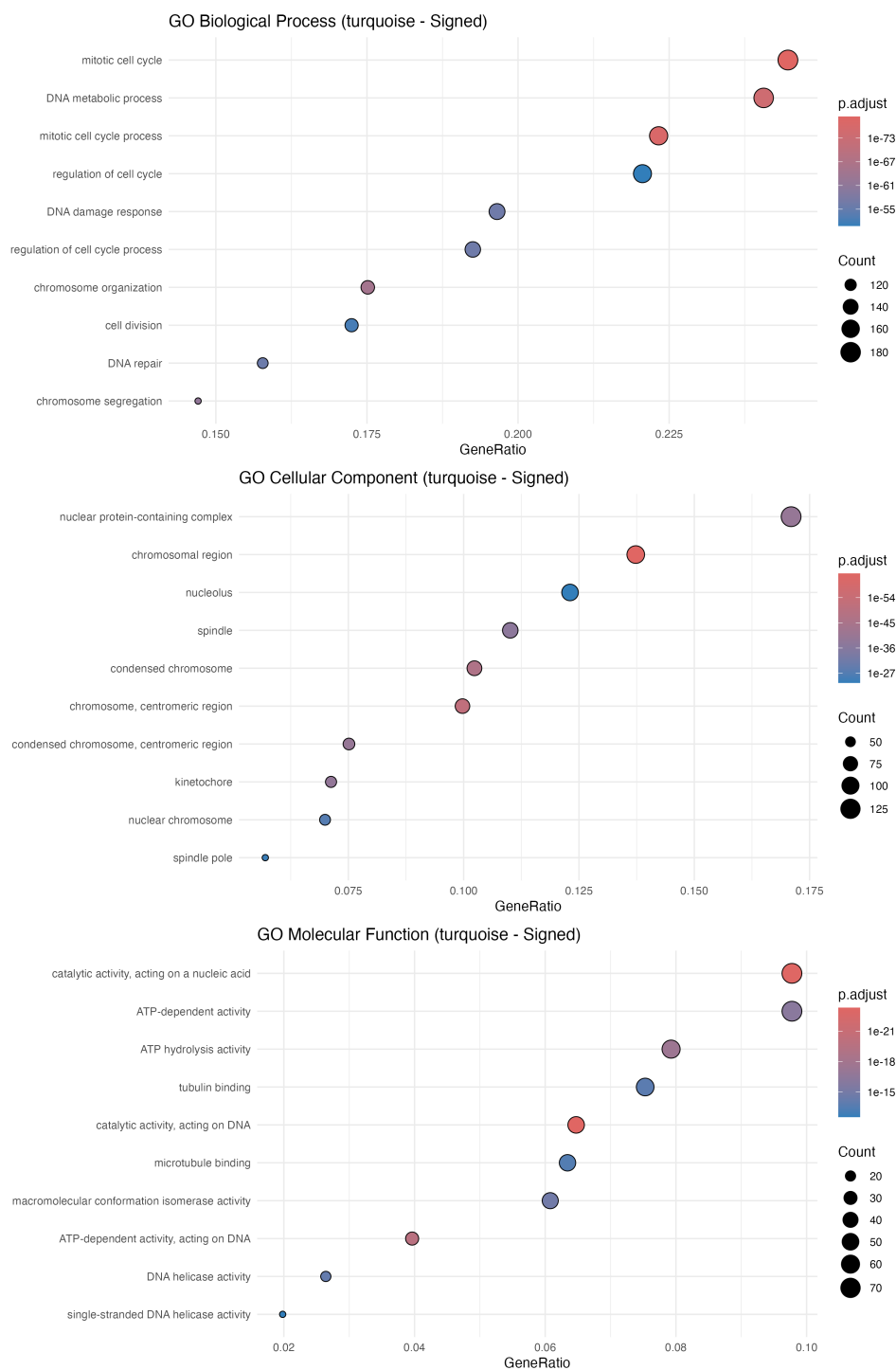


Figure 10: GO enrichment, signed turquoise module (793 genes).

turquoise is a **proliferation / cell-cycle** module: every top BP term is cell-cycle or DNA-metabolism related (mitotic cell cycle, DNA metabolic process, DNA damage response, chromosome segregation), CC terms are dominated by mitotic machinery (spindle, kinetochore, condensed chromosome, chromosomal region), and MF terms are dominated by DNA-acting catalytic/ATP-dependent activities (DNA helicase activity, catalytic activity on DNA/nucleic acid). This is fully consistent with turquoise's strong *positive* correlation with Basal ($r = 0.56$)

and strong *negative* correlation with LumA ($r = -0.73$): Basal is a highly proliferative PAM50 subtype, LumA the least. The module's top hub gene, *MCM10* (kME = 0.914), is a core component of the DNA replication machinery.

4.2 Brown: Ciliated/Differentiated Epithelial Identity

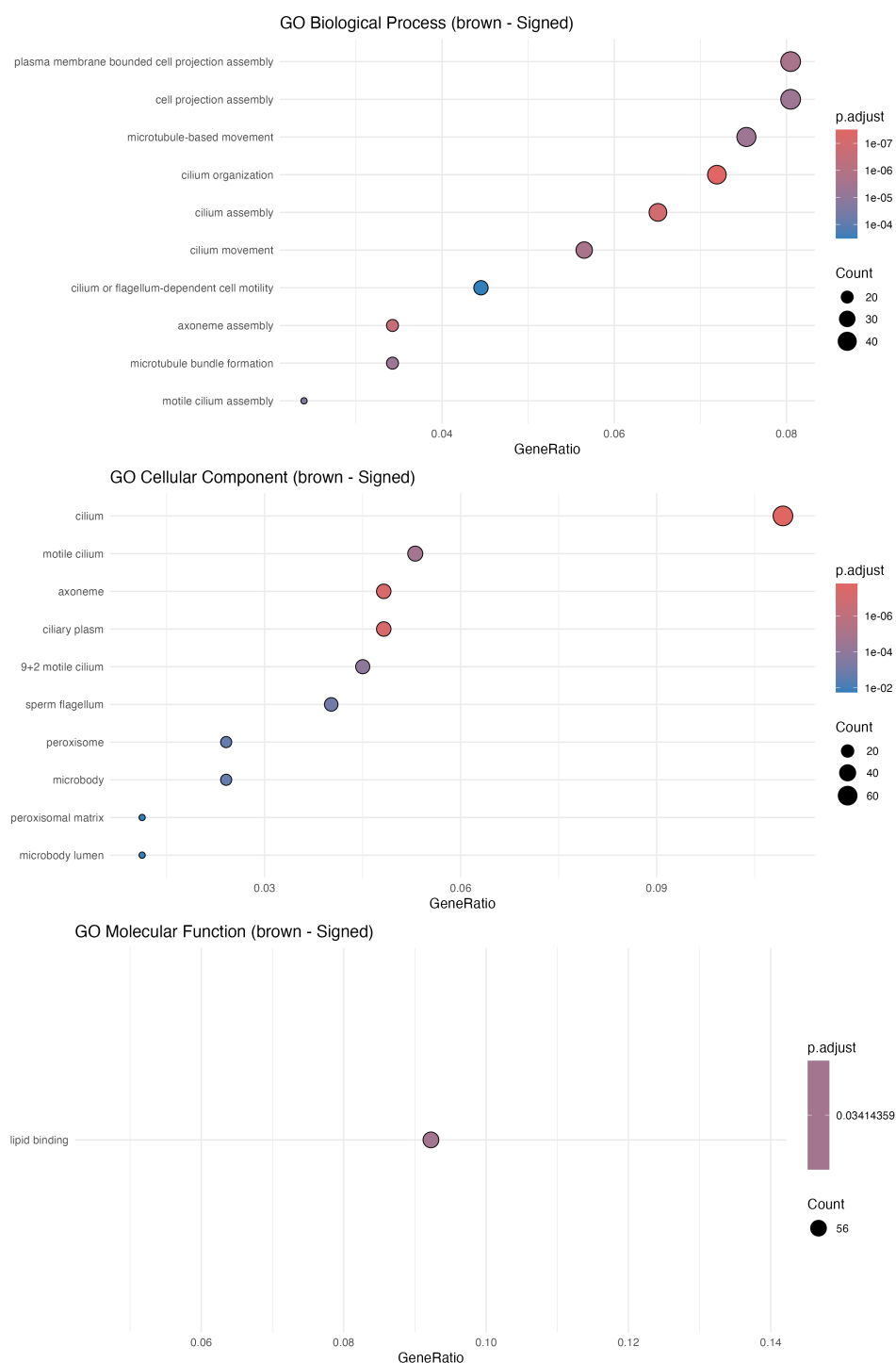


Figure 11: GO enrichment, signed **brown** module (**673** genes).

Cilium organisation/assembly/movement in BP, cilium/axoneme/motile cilium/9+2 motile cilium in CC, plus a peroxisome/lipid-binding signature. This is the strongest trait-correlated

module in the whole analysis (Basal $r = -0.86$, LumA $r = \mathbf{0.61}$), and its top hub gene is ***ESR1*** ($kME = 0.886$), the estrogen receptor gene that molecularly defines the luminal subtypes.

The module also contains *GATA3*, the most frequently altered hub gene in the entire table (11.9% mutated, 14.6% altered). Molecular Function returned only a single term, lipid binding, so the functional characterisation rests almost entirely on BP and CC.

4.3 Blue: Immune / Leukocyte Activation

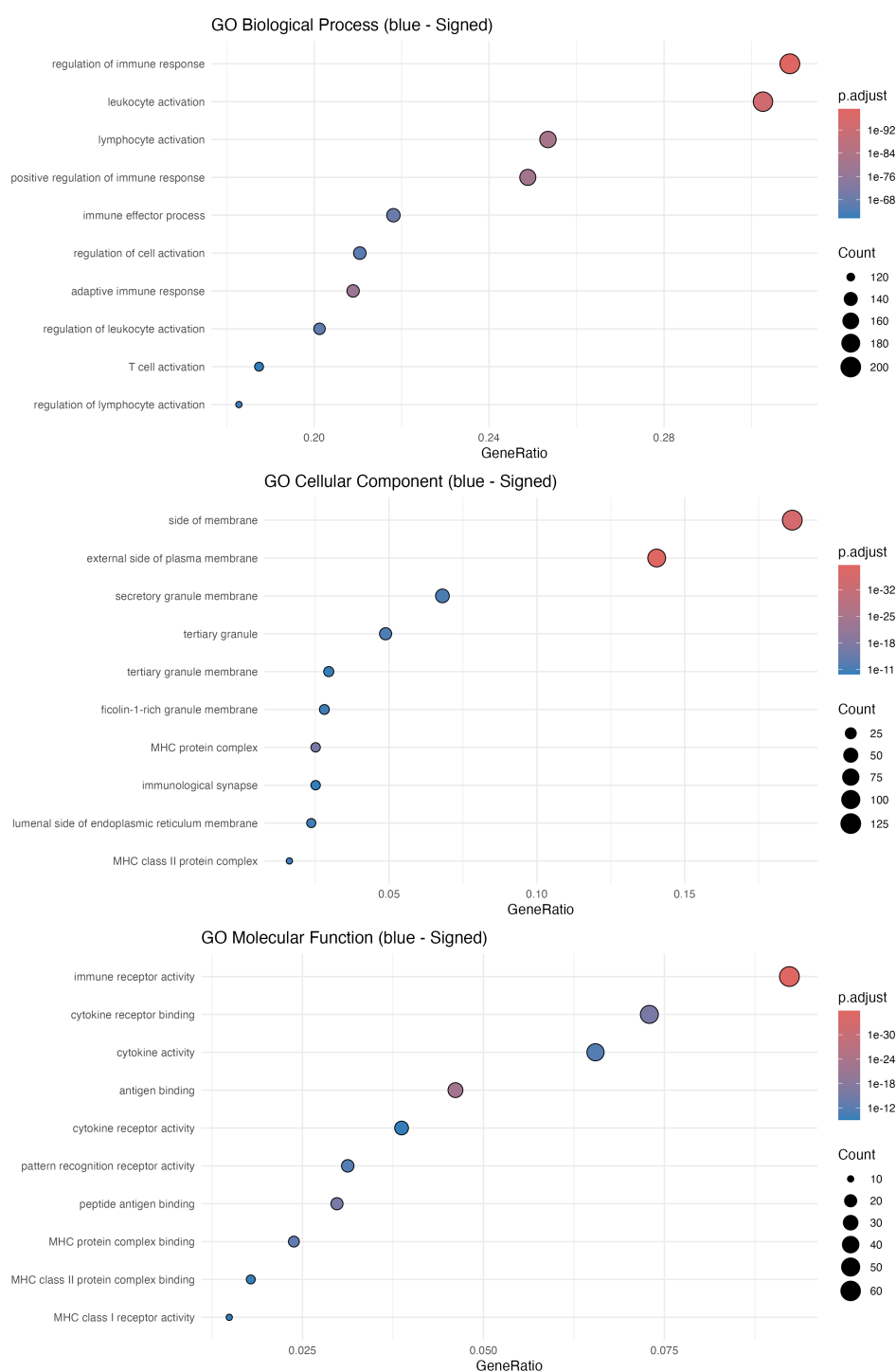


Figure 12: GO enrichment, signed blue module (699 genes).

blue is a clear **immune/inflammatory** module: BP is dominated by regulation of immune response, leukocyte and lymphocyte activation, T cell activation and adaptive immune response; CC includes side of membrane, external side of plasma membrane, MHC protein complex, immunological synapse and MHC class II protein complex; MF is dominated by immune receptor activity, cytokine receptor binding and activity, antigen and peptide-antigen binding, and MHC class I and class II binding specifically. This module shows only a weak positive correlation with Basal ($r = 0.24$) and weak negative correlations with LumA/LumB. The module's top hub gene is ***SNX20*** (**kME** = 0.958) , a sorting nexin expressed predominantly in immune cells.

4.4 Green: Ribosome Biogenesis / Oxidative Phosphorylation

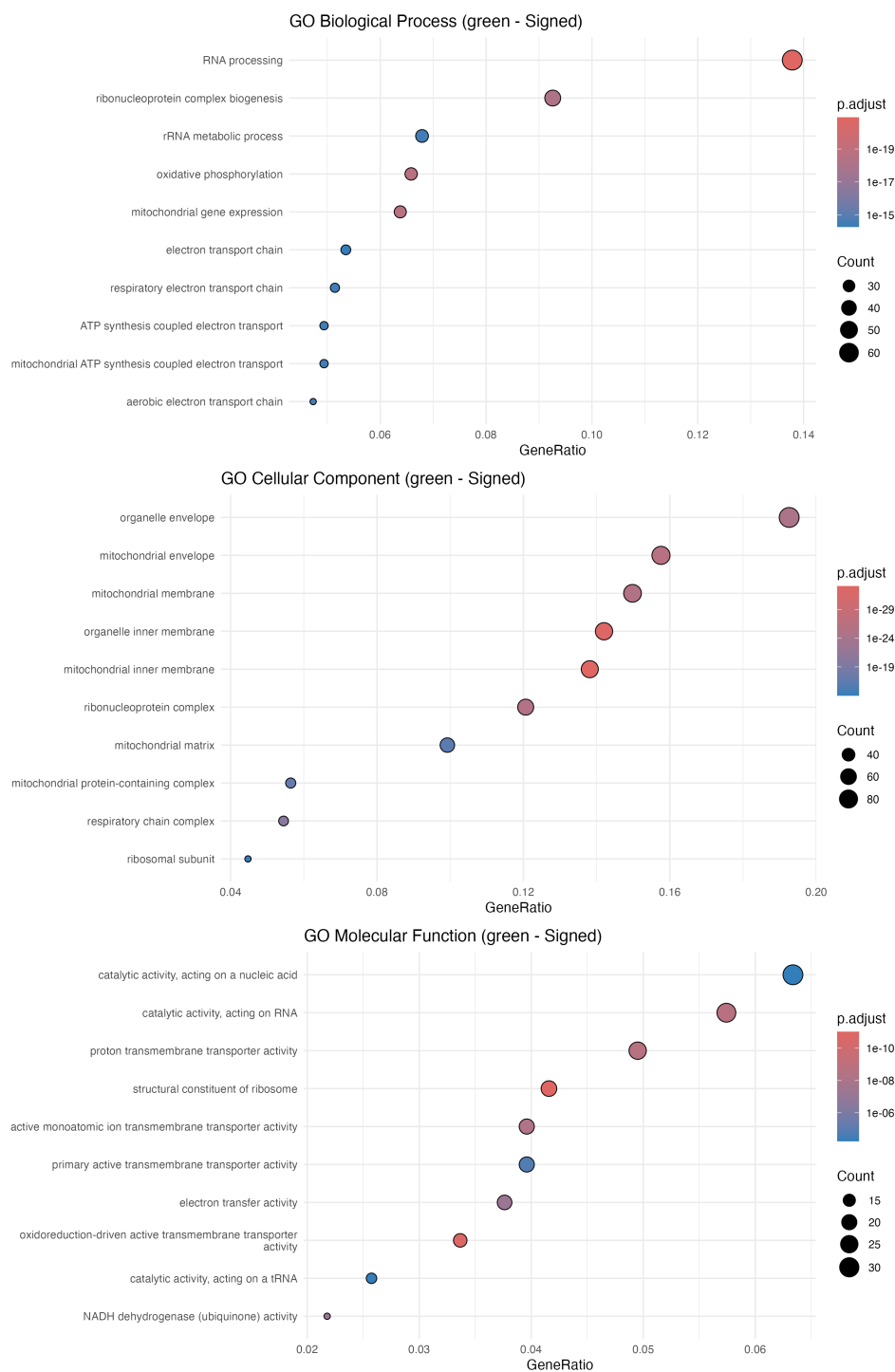


Figure 13: GO enrichment, signed **green** module (530 genes).

green shows a mixed but coherent **translational/metabolic** signature: RNA processing and ribonucleoprotein complex biogenesis dominate BP alongside oxidative phosphorylation and mitochondrial gene expression; CC is almost entirely mitochondrial (organelle/mitochondrial envelope, mitochondrial inner membrane, mitochondrial matrix) plus ribonucleoprotein complex and ribosomal subunit; MF is split between nucleic-acid catalytic activity/structural constituent of ribosome and transmembrane transporter/electron-transfer activities (consistent

with the electron transport chain machinery). **This module correlates positively with Basal ($r = 0.22$) and negatively with LumA ($r = -0.23$).** The more proliferative Basal tumours upregulate both ribosome biogenesis (to support high protein synthesis demand) and oxidative-phosphorylation/mitochondrial gene expression programmes together. The top hub gene, *NAA10* (**kME** = 0.858) , is an N-terminal acetyltransferase implicated in regulating protein synthesis and stability.

4.5 Yellow: Vasculature / Endothelium

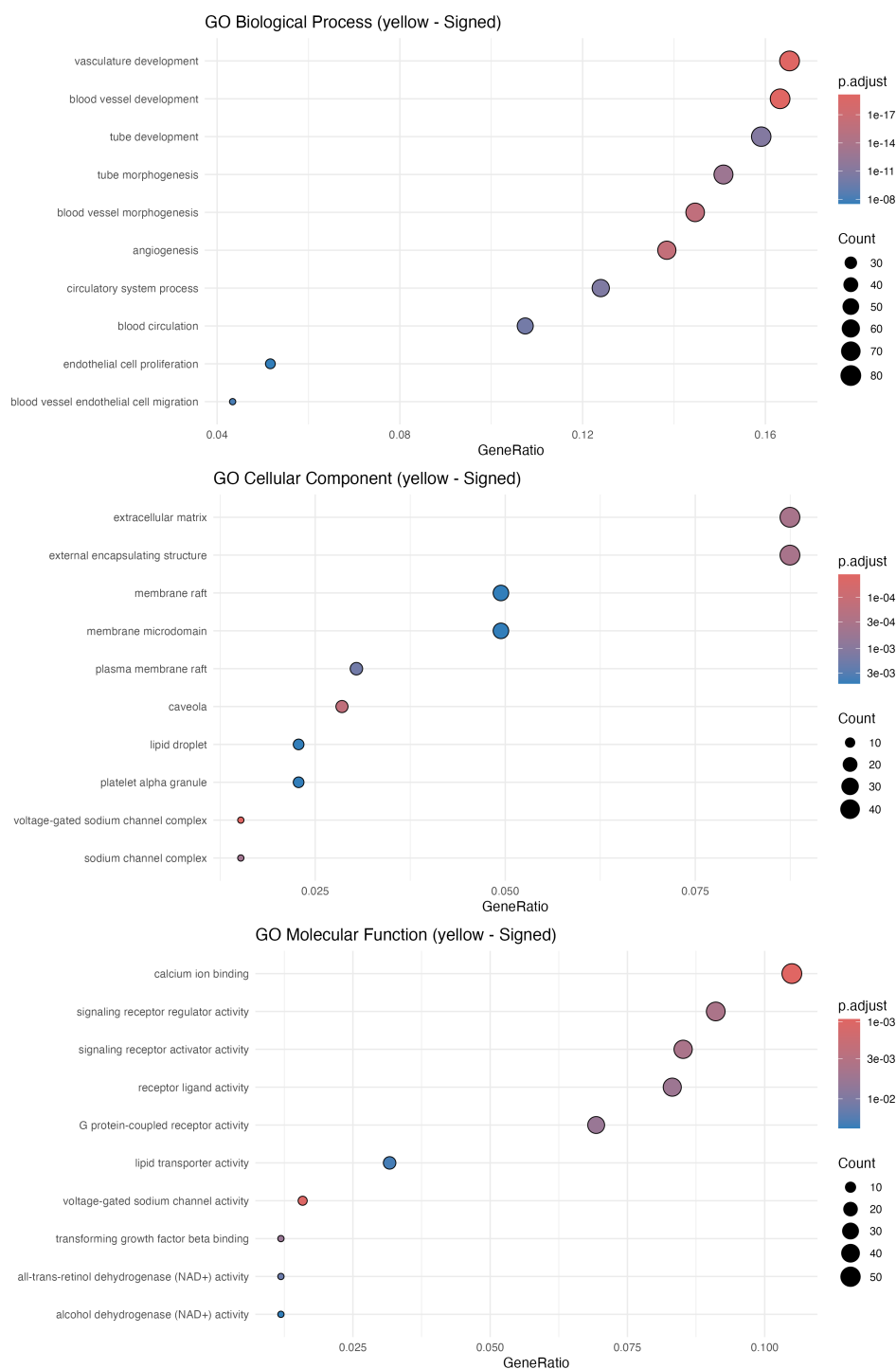


Figure 14: GO enrichment, signed yellow module (541 genes).

yellow's BP terms are dominated by **blood-vessel development and angiogenesis** (vasculature development, blood vessel development and morphogenesis, angiogenesis, circulatory system process, endothelial cell proliferation, blood vessel endothelial cell migration), with CC terms pointing to matrix and membrane microdomain structures (extracellular matrix, membrane raft, plasma membrane raft, caveola) and MF terms centred on calcium-ion binding and signalling receptor regulator and activator activity. This tracks yellow's positive correlation with LumA ($r = 0.51$) and negative correlations with Basal ($r = -0.26$) and LumB ($r = -0.31$). The top hub is now *S1PR1* (kME = 0.886), the sphingosine-1-phosphate receptor that governs endothelial barrier integrity

4.6 Red: Extracellular Matrix / Collagen

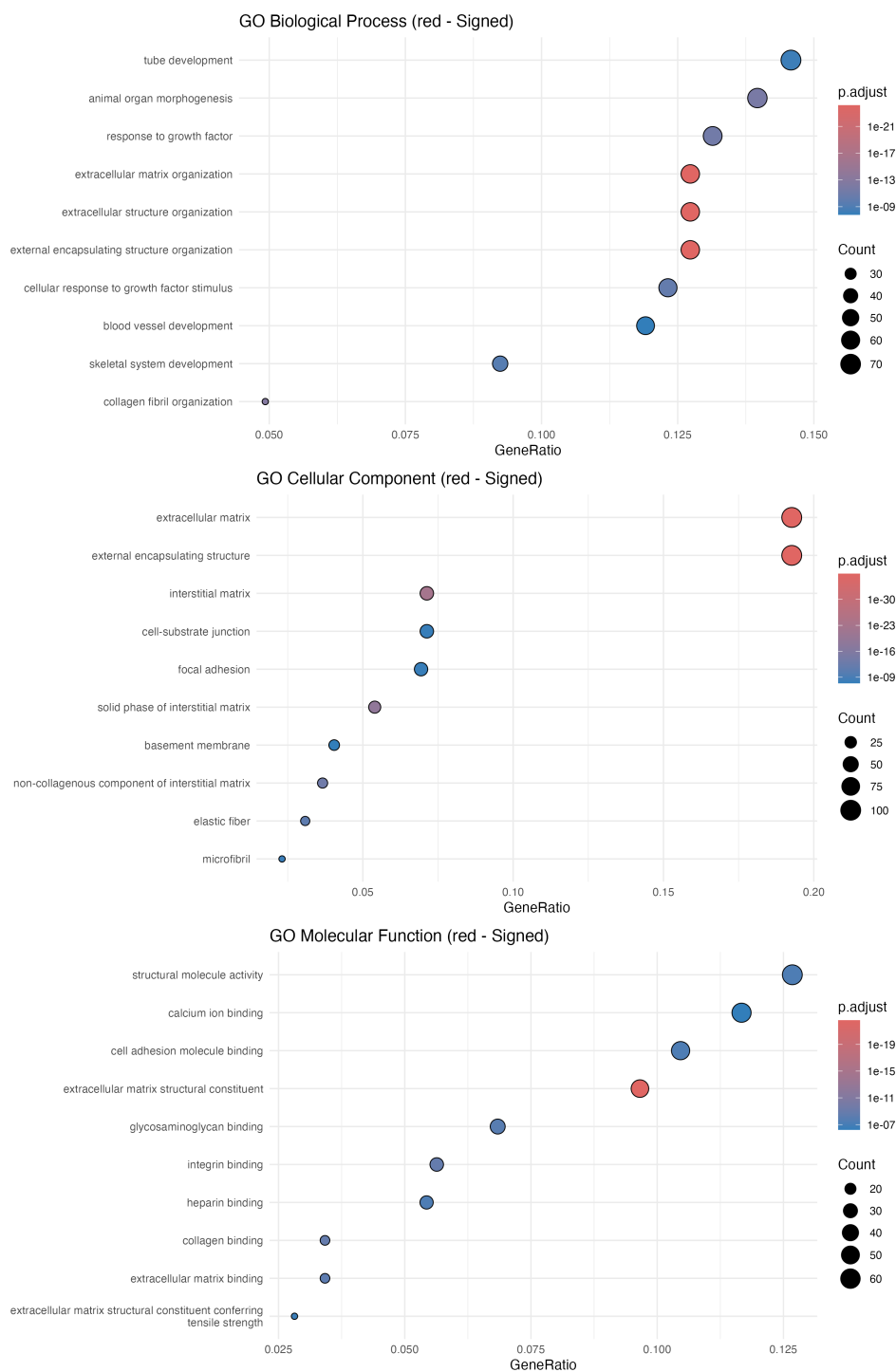


Figure 15: GO enrichment, signed red module (527 genes).

red is an **extracellular matrix / stromal** module, closely related to yellow but more specifically collagen-centred: BP terms include extracellular matrix and extracellular structure organisation, external encapsulating structure organisation, response to growth factor, and collagen fibril organisation; CC is dominated by extracellular matrix, interstitial matrix, basement membrane, elastic fibre and microfibril; MF is dominated by structural molecule activity, extracellular matrix structural constituent, collagen binding, integrin binding, heparin binding

and glycosaminoglycan binding. **Red correlates positively with LumA ($r = 0.30$) and negatively with Basal ($r = -0.20$) and LumB ($r = -0.19$)**, again pointing to differences in stromal/matrix composition across subtypes. The top hub is now *FBN1* (kME = 0.927), the fibrillin that nucleates elastic microfibrils, which lines up precisely with the elastic fibre and microfibril CC terms.

4.7 Pink: DNA Repair / Mitochondrial Envelope

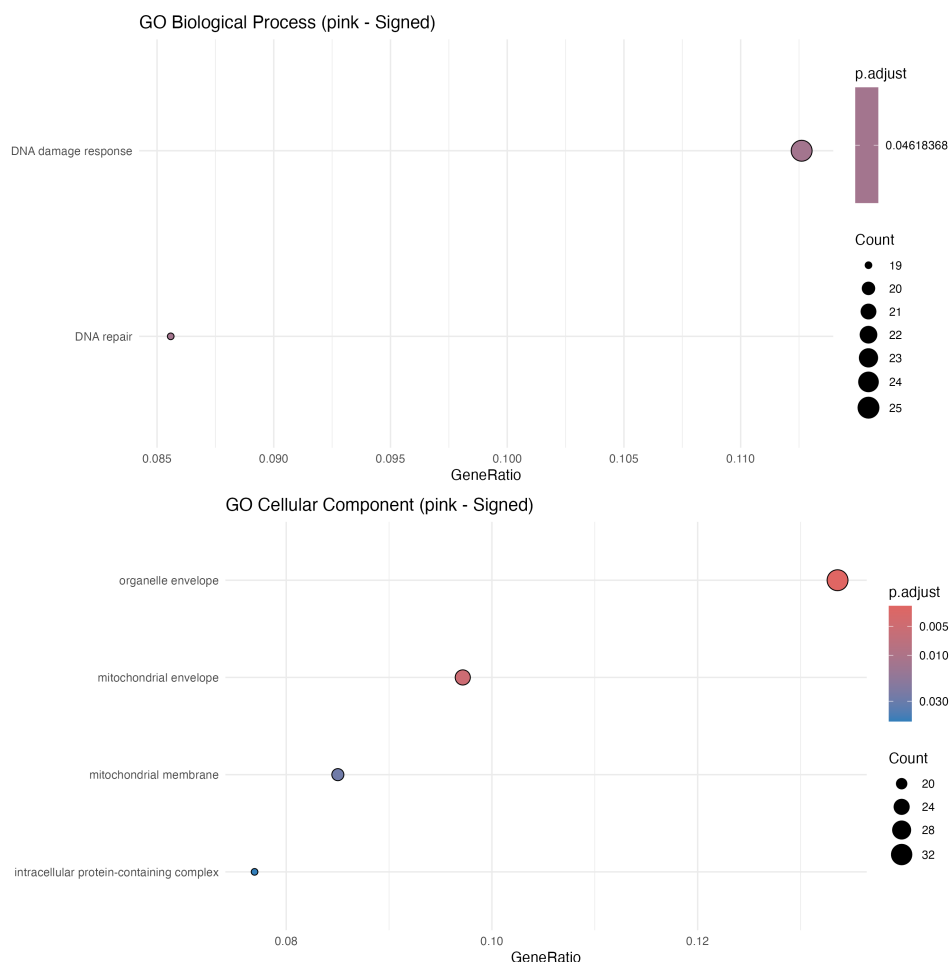


Figure 16: GO enrichment, signed pink module (268 genes). **Only BP and CC returned significant terms.**

pink returned only two Biological Process terms, DNA damage response and DNA repair, and four Cellular Component terms, all pointing to mitochondrial and organelle membranes (organelle envelope, mitochondrial envelope, mitochondrial membrane, intracellular protein-containing complex). Molecular Function returned nothing. The adjusted p -values are weak throughout, with the BP terms only just clearing the 0.05 threshold at $p_{adj} = 0.046$, so this module's characterisation is the least secure of the larger modules.

It correlates negatively with Basal ($r = -0.37$) and positively with LumA ($r = 0.24$). The top hub gene *MCRIP2* (kME = 0.893) is not itself a canonical mitochondrial or DNA-repair gene, although *NUBP2* and *CIAO3* immediately below it are both cytosolic iron-sulphur cluster assembly factors, a pathway that does link mitochondrial biology to DNA repair enzyme maturation.

4.8 Black: Ciliary Transition Zone

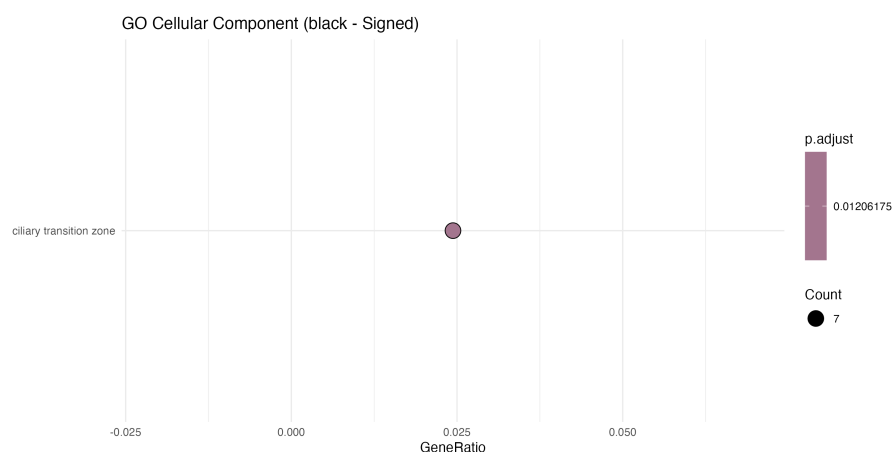


Figure 17: GO enrichment, signed **black** module (310 genes). **Only one term, in the CC ontology, reached significance.**

black is the most weakly characterised module in the analysis. Across all three ontologies it returned exactly one significant term, ciliary transition zone in Cellular Component, supported by 7 of its 310 genes at $p_{\text{adj}} = 0.012$. A single term backed by just seven genes is not a strong module in my opinion, and should be described as unresolved. It does carry a real trait signal, correlating negatively with Basal ($r = -0.39$) and positively with LumA ($r = 0.34$), so the genes are co-varying with subtype even though no annotated pathway explains why. The top hubs, *DMXL1*, *BLTP1*, *POLK* and *TTBK2*, span vesicle trafficking, lipid transport, translesion DNA synthesis and tau kinase activity with no obvious common thread; *TTBK2* is the only one with a documented ciliary role, as the kinase that initiates ciliogenesis at the distal appendage.

4.9 Magenta: Motile Cilia / Axonemal Dynein

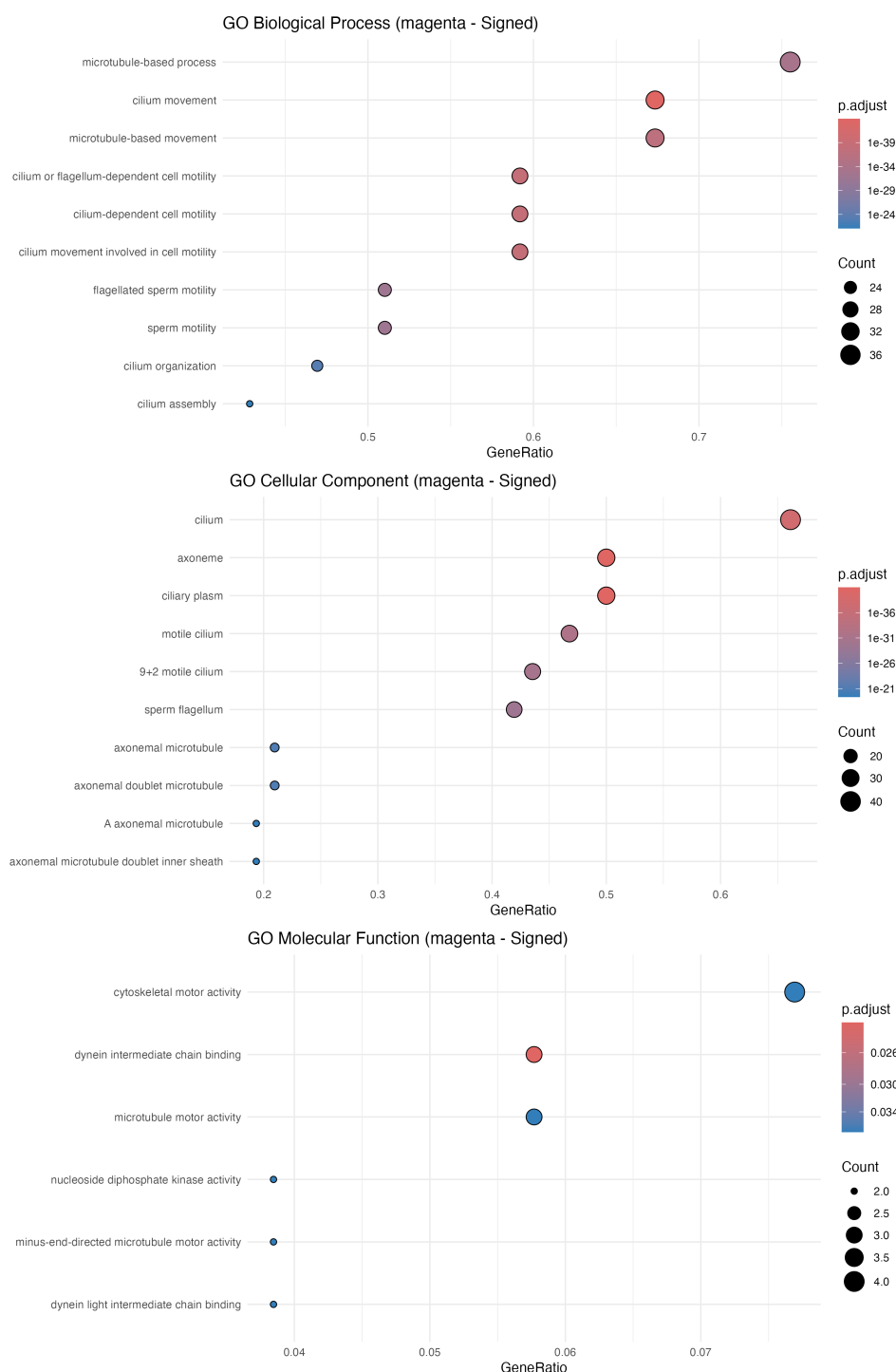


Figure 18: GO enrichment, signed **magenta** module (71 genes).

magenta is a well defined **motile cilium** module and, despite holding only 71 genes, it produces some of the strongest enrichment in the report. BP is entirely ciliary and microtubule-based (cilium movement, microtubule-based movement, cilium-dependent cell motility, flagellated sperm motility) with gene ratios above 0.5, meaning more than half the module maps to each of these terms; CC is dominated by cilium, axoneme, ciliary plasm, 9+2 motile cilium and axonemal doublet microtubule; MF picks up cytoskeletal and microtubule motor activity and dynein in-

intermediate chain binding. The adjusted p -values reach 10^{-39} in BP. It correlates negatively with Basal ($r = -0.29$) and positively with LumA ($r = 0.18$). The top hubs *ZBBX*, *WDR38*, *CFAP107*, *TEKT1* and *AKAP14* are all axonemal or ciliary structural proteins. **magenta** and **brown** both carry cilium signatures; **magenta** is specifically the motile, axonemal component, while **brown**'s ciliary terms sit alongside the luminal differentiation programme.

4.10 Purple: Chromosome 8q24.3 Amplicon

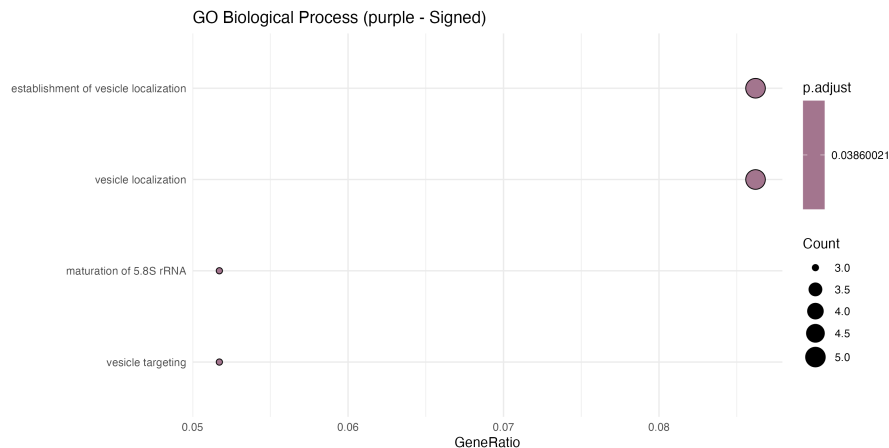


Figure 19: GO enrichment, signed **purple** module (67 genes). **Only the BP ontology returned significant terms.**

purple returned significant terms only in Biological Process, all related to vesicle localisation and targeting (establishment of vesicle localization, vesicle localization, vesicle targeting) plus one unrelated term (maturation of 5.8S rRNA), with very small gene counts per term (3 to 5 genes) and adjusted p -values only just below threshold at 0.039.

All twenty **purple** hub genes carry a TCGA-BRCA CNA frequency between 10.6% and 10.8%, an almost identical value, while their mutation frequencies are all under 1%. Genes with matched CNA frequencies are gained or lost as a block. *PUF60*, *HSF1*, *RECQL4*, *SHARPIN*, *SLC52A2*, *FBXL6*, *GPAA1*, *SCRIB*, *BOP1* and *VPS28* are all located at 8q24.3, a **recurrently amplified region in breast cancer**. Its trait profile (Basal $r = 0.15$, LumB $r = 0.20$, LumA $r = -0.26$) reflects the differing 8q24.3 amplification rate across subtypes.

4.11 Greenyellow: Histone Cluster / Chromatin Assembly

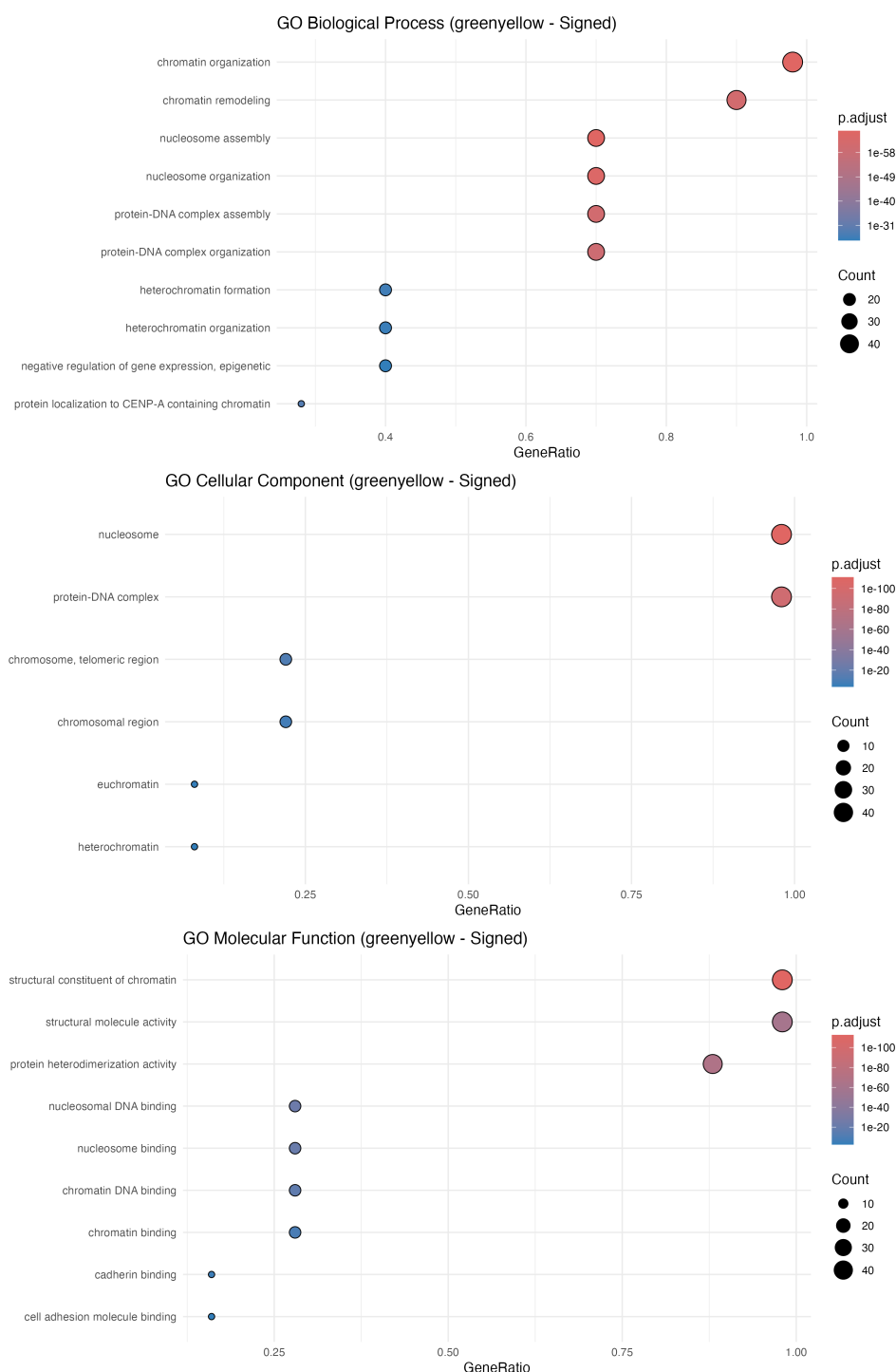


Figure 20: GO enrichment, signed **greenyellow** module (54 genes).

greenyellow is the smallest signed module at 54 genes and yet produces the single most extreme enrichment in the entire analysis. In Cellular Component, nucleosome and protein-DNA complex both reach a gene ratio close to 1.0 at p_{adj} below 10^{-100} , meaning essentially every gene in the module is a nucleosome component. BP is chromatin organisation, chromatin remodeling, nucleosome assembly and organisation, and protein-DNA complex assembly; MF is structural constituent of chromatin and protein heterodimerization activity. The top hubs

are *H4C4*, *H2BC17*, *H2BC7*, *H4C5*, *H2AC13* and *H3C10*, all replication-dependent clustered histone genes.

4.12 Summary

Table 6: One-line functional summary per signed module, its dominant PAM50 association, and its top hub gene. **Rebuilt entirely from the unfiltered run.**

Module	Dominant biology	Strongest subtype link	Top hub (kME)
turquoise	Cell cycle / DNA replication	Basal ↑ / LumA ↓	MCM10 (0.914)
brown	Luminal identity / ciliated epithelium	Basal ↓ / LumA ↑	ESR1 (0.886)
blue	Immune / leukocyte activation	Basal ↑ (weak)	SNX20 (0.958)
green	Ribosome biogenesis / OXPHOS	Basal ↑ / LumA ↓	NAA10 (0.858)
yellow	Vasculature / endothelium	LumA ↑ / LumB ↓	S1PR1 (0.886)
red	Extracellular matrix / collagen	LumA ↑ / Basal ↓	FBN1 (0.927)
pink	DNA repair / mitochondrial (weak)	Basal ↓ / LumA ↑	MCRIP2 (0.893)
black	Unresolved (one CC term only)	Basal ↓ / LumA ↑	DMXL1 (0.909)
magenta	Motile cilia / axonemal dynein	Basal ↓ (weak)	APOBEC4 (0.904)
purple	8q24.3 amplicon (positional)	LumA ↓ / LumB ↑	PUF60 (0.920)
greenyellow	Histone cluster / nucleosome	None significant	H4C4 (0.862)